Math 135 HW 6
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**1.**

(a)

$$y' = 2t(1+y), \qquad y(0) = 0.$$

This is linear:

$$y' - 2t\,y = 2t.$$

Integrating factor

$$\mu(t) = e^{\int -2t\,dt} = e^{-t^2}.$$

Then

$$\frac{d}{dt}\big(ye^{-t^2}\big) = 2te^{-t^2}.$$

Integrate:

$$y(t)e^{-t^2} = \int 2te^{-t^2}\,dt = -e^{-t^2} + C.$$

Thus

$$y(t) = -1 + Ce^{t^2}.$$

Using $y(0) = 0$ gives $0 = -1 + C$, so $C = 1$ and

$$\boxed{y(t) = e^{t^2} - 1.}$$

(b) Picard iterates with $y_0(t) \equiv 0$:

$$y_{n+1}(t) = y(0) + \int_0^t 2s\big(1 + y_n(s)\big)\,ds = \int_0^t 2s\big(1 + y_n(s)\big)\,ds.$$

$$y_0(t) = 0.$$

$$y_1(t) = \int_0^t 2s\,ds = t^2.$$

$$y_2(t) = \int_0^t 2s\big(1 + y_1(s)\big)\,ds = \int_0^t 2s(1+s^2)\,ds = \int_0^t (2s + 2s^3)\,ds = t^2 + \frac{t^4}{2}.$$

$$y_3(t) = \int_0^t 2s\big(1 + y_2(s)\big)\,ds = \int_0^t 2s\Big(1 + s^2 + \frac{s^4}{2}\Big)\,ds = \int_0^t (2s + 2s^3 + s^5)\,ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6}.$$

$$\begin{aligned} y_4(t) &= \int_0^t 2s\big(1 + y_3(s)\big)\,ds = \int_0^t 2s\Big(1 + s^2 + \frac{s^4}{2} + \frac{s^6}{6}\Big)\,ds \\ &= \int_0^t (2s + 2s^3 + s^5 + \tfrac{1}{3}s^7)\,ds = t^2 + \frac{t^4}{2} + \frac{t^6}{6} + \frac{t^8}{24}. \end{aligned}$$

**2.**

Here $y' = y$, $y(0) = 1$ and

$$y_{n+1}(t) = 1 + \int_0^t y_n(s)\,ds.$$

(a) $y_0(t) = e^t$.

$$y_1(t) = 1 + \int_0^t e^s\,ds = 1 + (e^t - 1) = e^t.$$

Then using $y_1$ again,

$$y_2(t) = 1 + \int_0^t e^s\,ds = e^t, \qquad y_3(t) = 1 + \int_0^t e^s\,ds = e^t.$$

(b) $y_0(t) = \cos t$.

$$y_1(t) = 1 + \int_0^t \cos s\,ds = 1 + (\sin t - \sin 0) = 1 + \sin t.$$

$$y_2(t) = 1 + \int_0^t (1 + \sin s)\,ds = 1 + \big(s - \cos s\big)\Big|_0^t = 1 + t - \cos t + 1 = t + 2 - \cos t.$$

$$y_3(t) = 1 + \int_0^t (s + 2 - \cos s)\,ds = 1 + \left(\frac{s^2}{2} + 2s - \sin s\right)\Big|_0^t = \frac{t^2}{2} + 2t + 1 - \sin t.$$

(c) $y_0(t) = (t+1)^2$.

Write $y_0(s) = s^2 + 2s + 1$.

$$y_1(t) = 1 + \int_0^t (s^2 + 2s + 1)\,ds = 1 + \left(\frac{s^3}{3} + s^2 + s\right)\Big|_0^t = 1 + t + t^2 + \frac{t^3}{3}.$$

$$y_2(t) = 1 + \int_0^t \left(1 + s + s^2 + \frac{s^3}{3}\right)ds = 1 + \left(s + \frac{s^2}{2} + \frac{s^3}{3} + \frac{s^4}{12}\right)\Big|_0^t$$

$$= 1 + t + \frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{12}.$$

$$y_3(t) = 1 + \int_0^t \left(1 + s + \frac{s^2}{2} + \frac{s^3}{3} + \frac{s^4}{12}\right)ds$$

$$= 1 + \left(s + \frac{s^2}{2} + \frac{s^3}{6} + \frac{s^4}{12} + \frac{s^5}{60}\right)\Big|_0^t = 1 + t + \frac{t^2}{2} + \frac{t^3}{6} + \frac{t^4}{12} + \frac{t^5}{60}.$$

**3.**

We have $f$ continuous and Lipschitz in $y$ with constant $L \geq 0$, and

$$y'(t) = f(t, y(t)), \qquad y(0) = y_0,$$

and Picard iterates

$$y_{n+1}(t) = y_0 + \int_0^t f(s, y_n(s))\, ds.$$

Also

$$y(t) = y_0 + \int_0^t f(s, y(s))\, ds.$$

Let $M = \|y - y_0\|_\infty = \sup_{t\in[0,a]} |y(t) - y_0(t)|$.

We show by induction that

$$|y(t) - y_n(t)| \le M\frac{(Lt)^n}{n!} \qquad (n \ge 0,\ t \in [0,a]).$$

Base $n = 0$:

$$|y(t) - y_0(t)| \le M = M\frac{(Lt)^0}{0!}.$$

Induction step: assume

$$|y(t) - y_n(t)| \le M\frac{(Lt)^n}{n!} \quad \text{for all } t \in [0,a].$$

Then

$$y(t) - y_{n+1}(t) = \int_0^t \big(f(s, y(s)) - f(s, y_n(s))\big)\, ds,$$

so

$$|y(t) - y_{n+1}(t)| \le \int_0^t L\,|y(s) - y_n(s)|\, ds \le \int_0^t L\,M\frac{(Ls)^n}{n!}\, ds.$$

Compute the integral:

$$|y(t) - y_{n+1}(t)| \le M\frac{L^{n+1}}{n!}\int_0^t s^n\, ds = M\frac{L^{n+1}}{n!}\cdot\frac{t^{n+1}}{n+1} = M\frac{(Lt)^{n+1}}{(n+1)!}.$$

Thus the inequality holds for $n + 1$, so by induction it holds for all $n$.

**4.**

We know $\|f_n - f\|_\infty \to 0$ on $[0,1]$. Then

$$\left|\int_0^1 f_n(t)\, dt - \int_0^1 f(t)\, dt\right| = \left|\int_0^1 (f_n(t) - f(t))\, dt\right| \le \int_0^1 |f_n(t) - f(t)|\, dt.$$

On $[0,1]$,

$$|f_n(t) - f(t)| \le \|f_n - f\|_\infty,$$

so

$$\left|\int_0^1 f_n(t)\, dt - \int_0^1 f(t)\, dt\right| \le \|f_n - f\|_\infty \cdot \int_0^1 1\, dt = \|f_n - f\|_\infty.$$

As $n \to \infty$, the right side goes to 0, so the difference of integrals goes to 0, i.e.

$$\int_0^1 f_n(t)\, dt \to \int_0^1 f(t)\, dt.$$

**5.**

Check $y(t) = 0$:

$$y'(t) = 0, \qquad 3y^{2/3} = 3 \cdot 0^{2/3} = 0,$$

so $y' = 3y^{2/3}$ and $y(0) = 0$.

Check $y(t) = t^3$:

$$y'(t) = 3t^2, \qquad y^{2/3}(t) = (t^3)^{2/3} = t^2,$$

so

$$3y^{2/3} = 3t^2 = y'(t),$$

and again $y(0) = 0$. So both are solutions.

The Picard–Lindelöf theorem requires $f(t, y) = 3y^{2/3}$ to be Lipschitz in $y$ near the initial point. Here

$$\frac{\partial f}{\partial y}(t, y) = 2y^{-1/3},$$

which is unbounded as $y \to 0$. Thus $f$ is *not* Lipschitz in any neighborhood of $y = 0$, so the hypotheses of Picard–Lindelöf fail. Therefore there is no contradiction.

**6.**

We have

$$y' = \frac{1}{y+1} = f(t, y), \qquad f(t, y) = \frac{1}{y+1}.$$

Here $f$ does not depend on $t$. We need $f$ and $\partial f/\partial y$ continuous near $(t_0, y_0)$.

$$\frac{\partial f}{\partial y}(t, y) = -\frac{1}{(y+1)^2}.$$

Both $f$ and $f_y$ are continuous whenever $y \neq -1$. Thus for any initial condition

$$y(t_0) = y_0 \quad \text{with } y_0 \neq -1,$$

Picard–Lindelöf guarantees a unique solution on some interval around $t_0$. If $y_0 = -1$ the theorem does not apply.

**7.**

We rewrite

$$ty' = y + 2t \quad \Rightarrow \quad y' = \frac{y + 2t}{t} = f(t, y), \qquad f(t, y) = \frac{1}{t}y + 2.$$

(a) The function $f(t, y)$ is defined and continuous only for $t \neq 0$, and

$$\frac{\partial f}{\partial y}(t, y) = \frac{1}{t}$$

is also continuous for $t \neq 0$. Thus for any $t_0 \neq 0$ the hypotheses of Picard–Lindelöf hold, so there is a unique solution on some interval around $t_0$. For $t_0 = 0$ the equation is not even defined, so no such guarantee holds.

(b) Fix $t_0 \neq 0$. The solution is guaranteed on any closed interval $[c, d]$ containing $t_0$ that stays away from $t = 0$.

If $t_0 > 0$, we must have $0 < c \leq t_0 \leq d$; any such interval $[c, d] \subset (0, \infty)$ works.

If $t_0 < 0$, we must have $c \leq t_0 \leq d < 0$; any such $[c, d] \subset (-\infty, 0)$ works.

On these intervals $f$ and $f_y$ are continuous, so there is a unique solution with domain $[c, d]$.